



INDIA.RUNS

Build what next India runs on

Team Name : Limit Breakers

Team Leader Name : Atchyuta Pavan Karthikeya

Problem Statement : Recruiters often miss highly relevant candidates because traditional hiring systems rely heavily on keyword matching rather than understanding true candidate fit.

Solution Overview

What is your proposed solution?

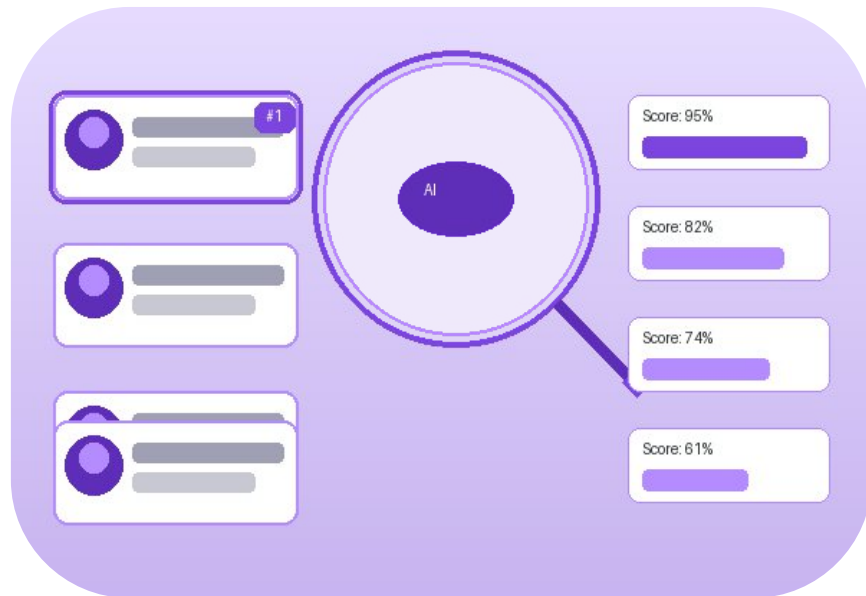
An explainable AI-powered candidate ranking system.

Our system evaluates candidates using:

- **What they know** (skills & experience)
- **What they have done** (career evidence)
- **How they behave** (activity & recruiter signals)
- **Why they fit the role** (explainable ranking)

What differentiates your approach?

- Goes beyond keyword matching
- Uses behavioral hiring signals
- Detects suspicious profiles and honeypots
- Provides recruiter-friendly explanations
- Produces trusted, explainable rankings



JD Understanding & Candidate Evaluation

Key JD Requirements

- Search & Retrieval Systems • Ranking Systems • Recommendation Engines
- Production ML Experience • Python Expertise • Product Mindset

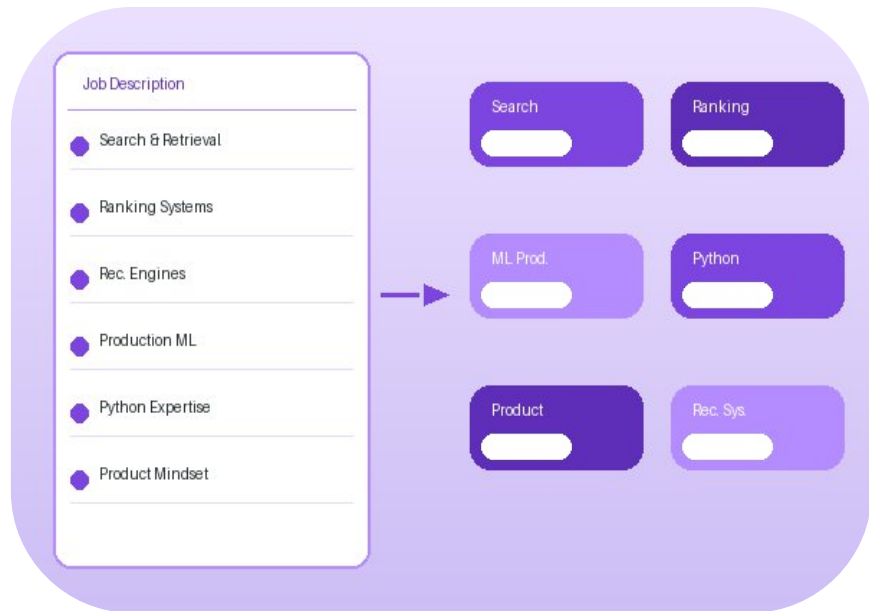
Most Important Candidate Signals

Technical: Retrieval/Search Experience, Ranking Systems, Recommendation Systems, Production Deployments

Behavioral: Recruiter Response Rate, Activity Recency, Interview Completion, Open-to-Work Status

Career: Ownership, Startup Experience, Product Impact

Rather than matching keywords, the system evaluates whether a candidate has demonstrated the capabilities required for the role.



Ranking Methodology

Ranking Pipeline

Parse JD requirements → Extract candidate features →
Compute semantic relevance → Score behavioral signals
→ Score career evidence → Detect honeypots →
Generate final ranking

Models, Algorithms & Heuristics

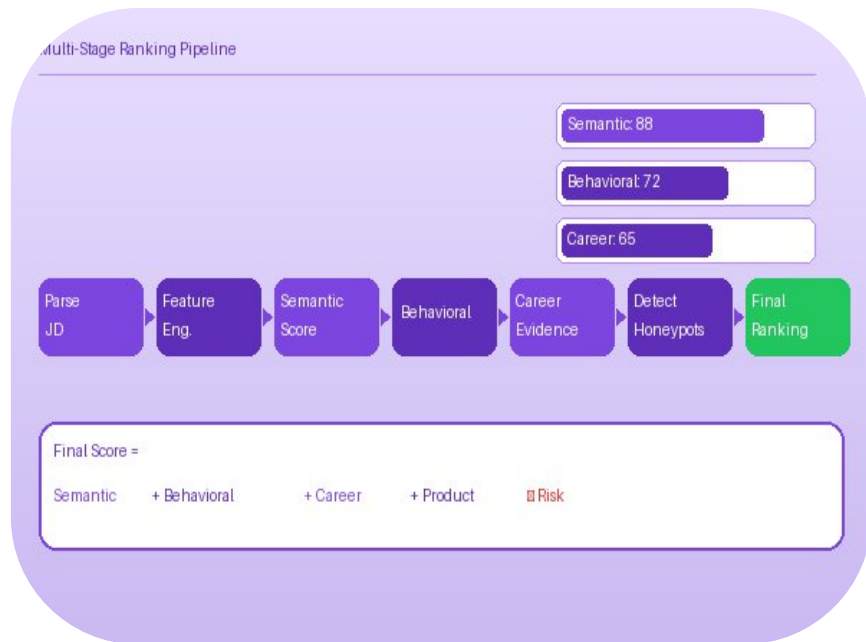
Semantic Similarity Scoring • Feature Engineering • Multi-
Stage Ranking Pipeline • Behavioral Signal Weighting •
Honeypot Detection Rules

Final Score Formula

Final Score = Semantic Relevance + Behavioral Intelligence +
Career Evidence + Product Fit – Risk Penalties

The ranking score combines:

- **Technical relevance** — does the candidate know the subject?
- **Career evidence** — have they done it before?
- **Behavioral signals** — are they actively engaged and reachable?
- **Profile quality checks** — is the profile credible and consistent?



Explainability & Data Validation

How Ranking Decisions Are Explained

Each candidate receives: Overall Score • Positive Factors • Negative Factors • Skill Alignment • Career Highlights • Behavioral Highlights

Preventing Hallucinations

Evidence-based explanations • Only profile-derived insights • No unsupported generated claims

Handling Suspicious Profiles

Timeline Validation • Contradiction Detection • Keyword Stuffing Checks • Honeypot Detection • Risk Penalties

Every recommendation includes both strengths and risk factors so recruiters can understand why a candidate was ranked.

Sample Explainability Output

Candidate #18472 | Final Score: 94.2

Top Strengths

- ✓ Retrieval Systems experience
- ✓ Ranking Infrastructure background
- ✓ Product Experience (2 shipped products)
- ✓ High Recruiter Engagement score

Risk Factors

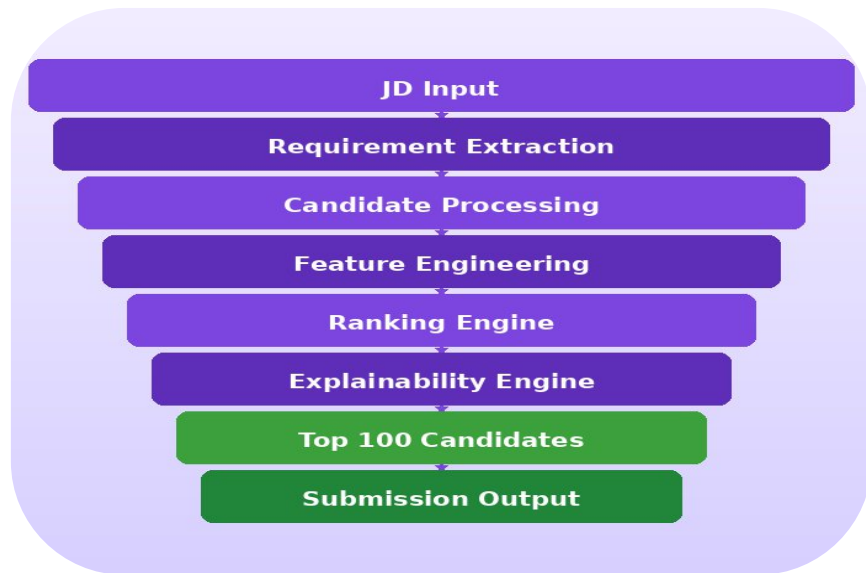
⚠ Longer notice period (60 days)

Every ranking decision is fully explainable.

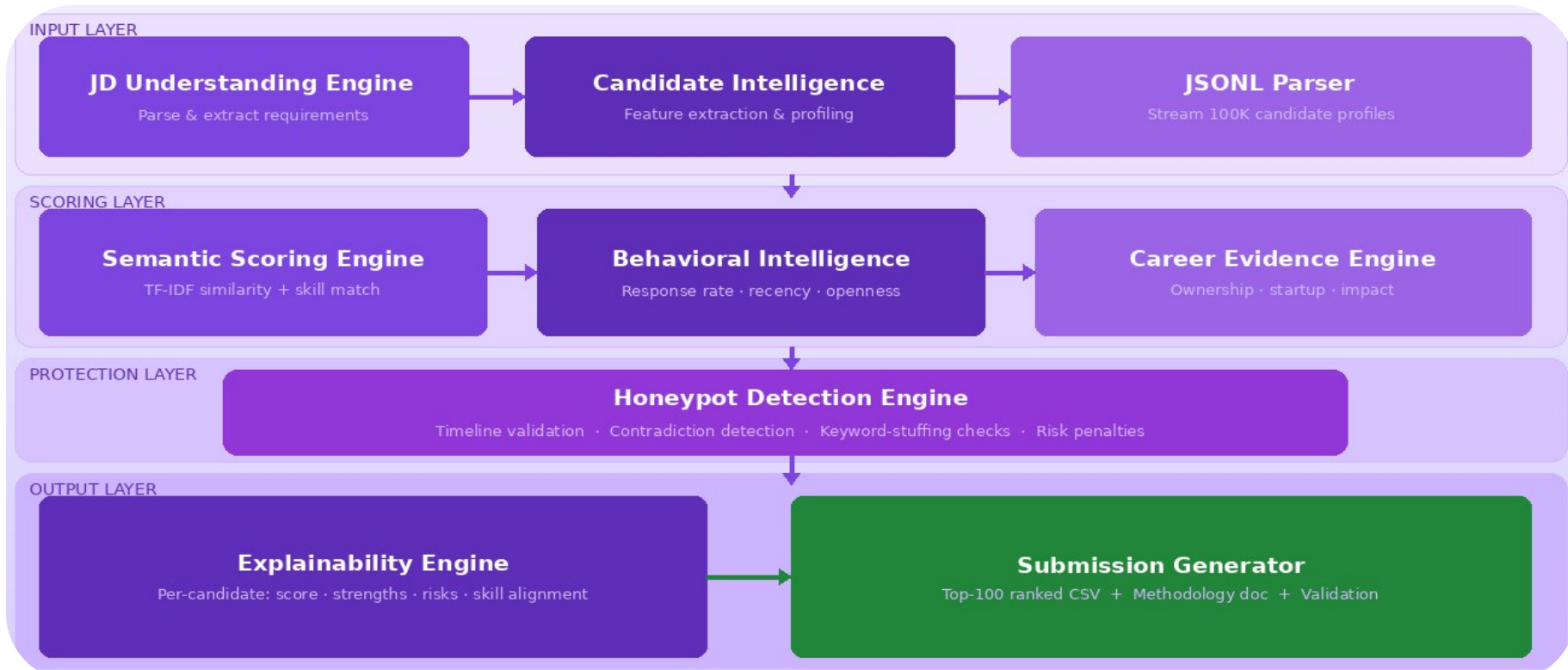
End-to-End Workflow

Complete Workflow: JD Input to Ranked Output

- ① JD Input → ② Requirement Extraction → ③ Candidate Processing
④ Feature Engineering → ⑤ Ranking Engine → ⑥ Explainability Engine
⑦ Top 100 Candidates → ⑧ Submission Output



System Architecture



The system uses a modular architecture where candidate understanding, semantic scoring, behavioral intelligence, validation, and explainability operate independently to ensure scalability, transparency, and maintainability.

Results & Performance

Ranking Quality Insights

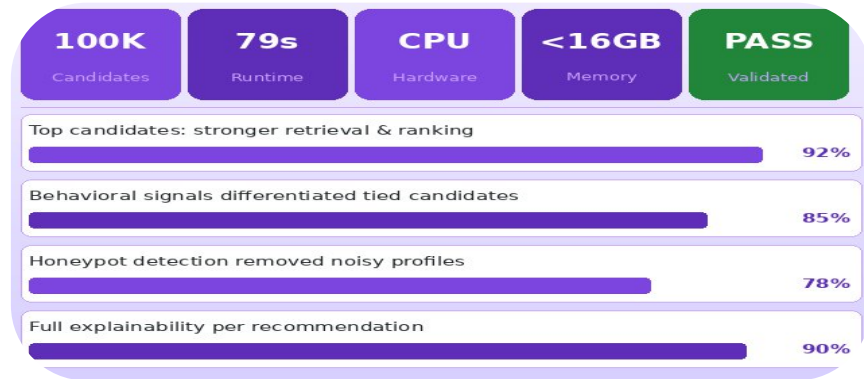
- Top-ranked candidates consistently showed stronger retrieval, ranking, and recommendation-system experience than lower-ranked ones.
- Behavioral signals helped distinguish technically similar candidates where skill overlap alone was insufficient.
- Honeypot detection removed suspicious profiles from the final shortlist, reducing noise for recruiters.
- Explainability enabled transparent recruiter review — every recommendation included positive factors, risks, and skill alignment.

Runtime & Constraints

Metric	Result
Candidates Processed	100,000
Runtime	79 seconds
Hardware	CPU Only
Memory Usage	<16 GB
Submission Validation	Passed ✓

Business Impact

- Reduces dependence on keyword matching
- Surfaces hidden but relevant candidates
- Improves recruiter trust through explainability



Key Outcomes

- OK 100K candidates processed in 79 sec on CPU
- OK Explainable rankings for every candidate
- OK Suspicious profiles filtered — honeypots removed
- OK Validation-compliant submission passed

Technologies Used

Technology Why We Used It

- **Python** Rapid AI system development; rich ecosystem for NLP and data pipelines
- **Pandas** Efficient structured processing of 100,000 candidate profiles at scale
- **NumPy** Fast vectorised numerical scoring across all candidate signals
- **JSONL Processing** Streaming candidate parsing without loading all 100K records into memory
- **Modular Pipeline** Each engine is independently testable — supports explainability and maintainability

No external ML APIs. No cloud dependencies. Fully self-contained and reproducible.

Submission Assets

Deliverables

✓ GitHub Repository ✓ Ranked Candidates CSV ✓ Explainability Report ✓ PPT Submission ✓ Validation-Compliant Output

Repository

github.com/pavankarthikeyaatchyuta-lab/Redrob-Intelligent-Candidate-Ranker

“We didn’t build a keyword matcher. We built an explainable AI recruiter that understands candidate relevance, hiring readiness, and recruiter intent.”



INDIA.RUNS

Build what next India runs on

THANK YOU